

Practice Quiz: Body Science — Module 06: Learning

Name: ____ Date: ____

Part A: Multiple Choice

1. When your immune system encounters a new virus and then "remembers" how to fight it next time, this is an example of:

A) A coincidence B) Learning at the cellular level — your immune system updates its model to predict and respond to the virus faster C) Your immune system guessing randomly D) Something that only the brain can do

1. Neuroplasticity means:

A) Your brain is made of plastic B) Your brain physically changes its connections based on experience — it rewires itself as it learns C) Your brain stops changing after age 5 D) Learning doesn't change the brain at all

1. Why does sleep help you learn?

A) It doesn't — sleep is just rest B) During sleep, your brain replays experiences and strengthens the neural connections for what you learned, updating its models C) You learn new things while dreaming D) Sleep erases what you learned during the day

1. "Use it or lose it" in the brain means:

A) You should exercise every day B) Neural pathways that get used frequently get stronger, while those that aren't used get weaker — your brain prunes connections that aren't making useful predictions C) You lose brain cells every day no matter what D) This only applies to muscles, not the brain

1. A baby learning to walk falls down hundreds of times. Each fall is:

A) A failure that should make the baby give up B) A prediction error that gives the baby's brain data to improve its balance and movement models C) Proof that babies aren't meant to walk D) Random and teaches nothing

1. When you practice a sport and the movements start to feel automatic, your brain has:

A) Stopped learning B) Moved the skill from conscious processing to a more efficient, automatic system — freeing up mental resources for new challenges C) Deleted the information D) Run out of storage space

1. Stress hormones (like cortisol) can hurt your ability to learn because:

A) Stress has no effect on the brain B) High cortisol can interfere with the brain's ability to form new connections and update models C) Stress makes you smarter D) Cortisol only affects your muscles

Part B: Short Answer

1. Explain how your **immune system learns** from a vaccination. What is the "prediction error"? How does the immune system "update its model"? How does this help it respond faster to the real virus?
2. Describe how **neuroplasticity** works using a real example (learning an instrument, a language, or a sport). What is happening in your brain as you practice? Why does it get easier over time?
3. A student stays up all night to cram for a test. Using what you know about how the brain learns during **sleep**, explain why this strategy is less effective than studying for shorter periods over several days with good sleep in between.